SageMath: A First Look at Public-Key Cryptography

In the last article in this SageMath series (published in the March 2025 issue of *Open Source For You*), we concluded our discussion on symmetric-key encryption. As promised, this article will focus on public-key cryptography (PKC). However, before diving into that, we will briefly examine some recent developments in the field of quantum computing. We will also introduce the concept of cryptanalysis to understand how these emerging trends could have potentially catastrophic implications for cybersecurity.

A few months ago, I had a déjà vu moment when Microsoft announced the development of a quantum computing chip. It reminded me of an earlier experience: while I was writing a series of articles on artificial intelligence for *OSFY*, something historic happened. On 30th November 2022, ChatGPT was made publicly available. I started experimenting with it immediately and managed to publish an article featuring ChatGPT in the January 2023 issue of *OSFY*. I still proudly claim — though I have no concrete proof — that this was the first mention of ChatGPT in a printed magazine in India.

Fast forward to the present: I am now writing a series on SageMath and currently focusing on cybersecurity. And right on cue, along comes an invention that could potentially upend many foundational aspects of cybersecurity as we know it.

Majorana 1: A milestone in quantum computing?

Microsoft's new quantum chip, named Majorana 1, is being hailed as a breakthrough. The chip is said to support quantum superposition and entanglement at room temperature, making room-temperature qubits a practical reality — something that, until now, seemed decades away.

The chip is named after Ettore Majorana, a brilliant Italian physicist often described as "the next Einstein." Majorana mysteriously disappeared in 1938 at the age of 31, and his legacy includes the theoretical concept of Majorana fermions — particles that are their own antiparticles. Microsoft's approach to topological quantum computing is believed to leverage ideas inspired by these fermions, which are thought to provide enhanced stability against quantum decoherence — one of the key challenges in building scalable quantum systems.

However, much like the elusive scientist himself, the Majorana 1 chip has sparked a fair amount of scepticism in the scientific community. The claims are bold, and peer-reviewed validation is still awaited. Still, if the results hold up, this could dramatically accelerate the timeline for practical quantum computing — from centuries or decades down to just a few years.